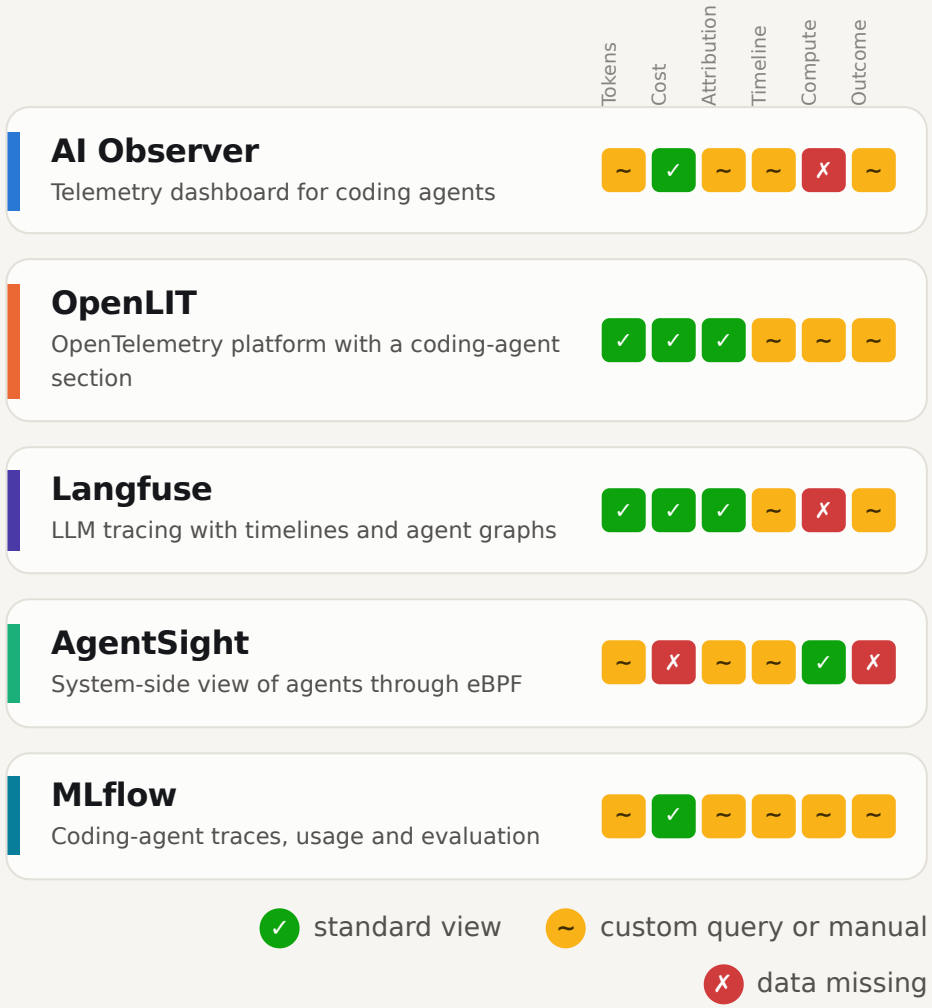
Five tools. What each one actually shows.

The shortlist, described from real interfaces: anonymized public examples, project documentation and a running AI Observer. For each tool: what you see, how our goals map onto it, and what is missing.



00 · WHAT WE LOOK FOR

Can a person see the waste without reading raw logs?

S1

Rerun without a change

Tests or other commands rerun with no code or environment change.

compute time

S2

Expensive waiting

A lead agent keeps calling its model while another agent works.

tokens

S3

Polling

Frequent process polls or status messages without new information.

tokens time

S4

Long wait

A wait or blocker on the path to delivery.

time

S5

Spend, no result

High resource use without a successful outcome.

tokens compute time outcome

Ordinary environments
Codex, Claude Code and OpenCode in a plain devcontainer or terminal, without Subyard.

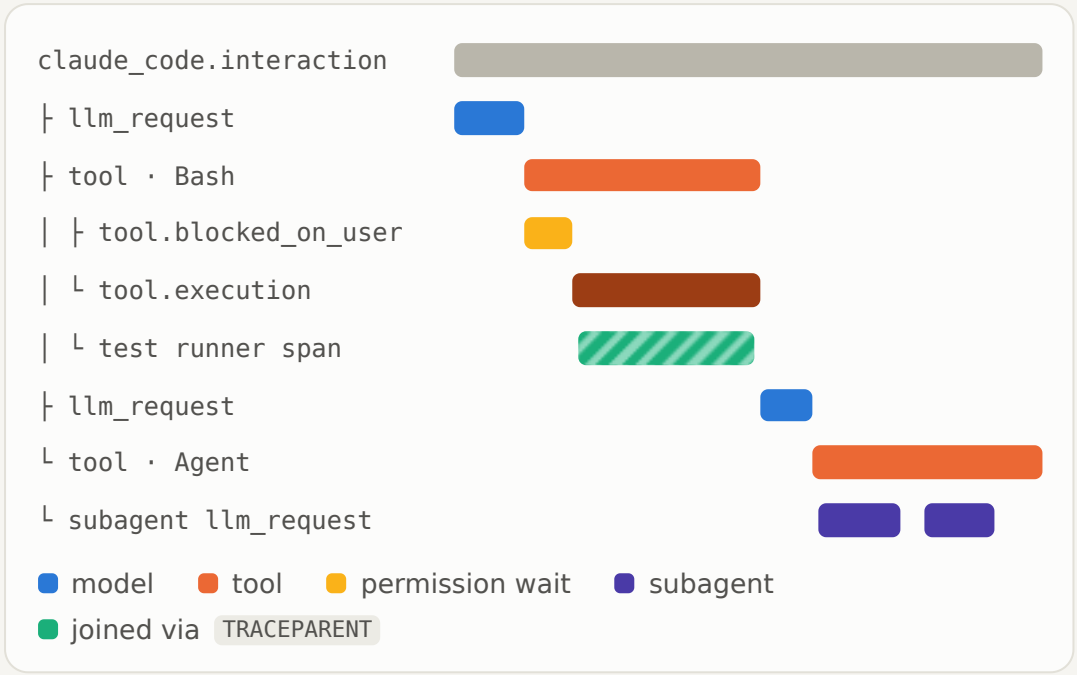
Tokens, counted once
Input, cached input, output and reasoning per model; cached is part of input.

Compute and time
CPU, memory and processes, including test runs; a timeline of work, waits and blockers.

Attribution
From task to agent or subagent to the step or tool call. Cost estimate and outcome are should-haves.

Every rating in this deck uses one scale: ✓ visible in the tool's standard views · ~ needs a custom query or manual correlation · ✗ the data is missing.

The agents already emit most of the model side.



Span names from the Claude Code monitoring docs. Bar lengths are illustrative, not measured.

Claude Code

- Traces (beta) with `CLAUDE_CODE_ENHANCED_TELEMETRY_BETA=1`; subagent spans carry `agent_id` and `parent_agent_id`.
- Bash subprocesses get `TRACEPARENT`, so a test runner can join the trace.
- Metric `token.usage` by type: input, cacheRead, cacheCreation, output.

Codex CLI

- Opt-in `[otel]`: events for API requests, prompts, tool approvals and results.
- Metric `turn.token_usage`: input, cached_input, output, reasoning_output.

OpenCode

- Telemetry through plugins: third-party OTLP plugin, or Langfuse's.

IMPLICATION **Configuration covers tokens, cost and tool calls. None of the agents reports CPU, memory or the processes it starts.**

00 · WHERE THE DATA COMES FROM

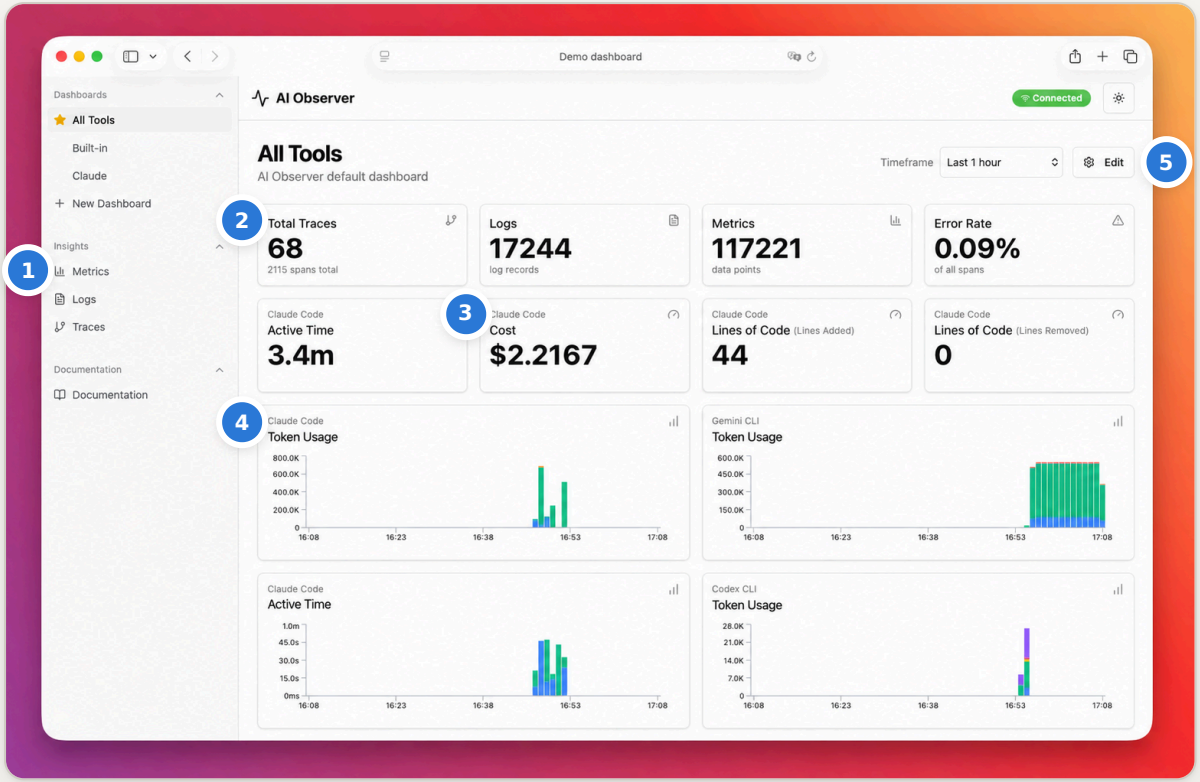
The data source decides what a tool can see.

	AGENT OTLP metrics, logs, traces	SESSION FILES ~/.claude, ~/.codex	AGENT HOOKS plugins, Stop hook	KERNEL EBPF TLS, exec, files	HOST AND PROCESSES /proc, collector	WHAT IT CAN SEE
AI Observer	● serve mode	● watch mode	—	—	—	Model calls, tokens, cost, tool calls. The two modes exclude each other.
OpenLIT	●	● token deltas	● openlit coding install	—	● separate collector	Turns, tools, subagents, Git outcome; host and process metrics beside them.
Langfuse	● traces only	● read by plugins	● Claude Code, Codex, OpenCode	—	—	The agent's own steps in detail; nothing outside the agent.
AgentSight	— exports only	● without root	—	● root or CAP_BPF	●	Processes, files, LLM traffic and CPU of any command; not the agent's intent.
MLflow	● traces only	● read by hooks	● all three agents	—	● host metrics, separate run	Tool/model spans, usage and assessments; no test-process attribution.

PATTERN **Four tools trace the agent's own activity. Per-process compute needs system-side observation.**

01 · AI OBSERVER 0.6.0 INTERFACE

A telemetry dashboard for coding agents.



Dashboard. Public AI Observer repository example, anonymized.

1 **Navigation**
Dashboards, Metrics, Logs, Traces, Sessions, built-in docs.

2 **Built-in tiles**
Traces, logs, metric points, error rate, active services.

3 **Metric Value widget**
One number per service: active time, cost, lines of code.

4 **Metric Chart widget**
Token usage over time, stacked by Cache Creation, Cache Read, Input and Output.

5 **Edit**
Add, move and remove widgets; several saved dashboards.

\+ **Sessions page**
Agent, start, duration, model, message count. A transcript opens as Chat or Timeline: a chronological list of prompts, tool calls and results.

\+ **Traces page**
Span waterfall, for example Codex `run_task → run_turn → reasoning`. Empty without OTLP traces.

Seen on a running instance: every page was checked on 0.6.0 in Subyard (watch mode, Claude Code and Codex). Its screenshots are not published because they contain session data.

01 · AI OBSERVER FIT TO OUR GOALS

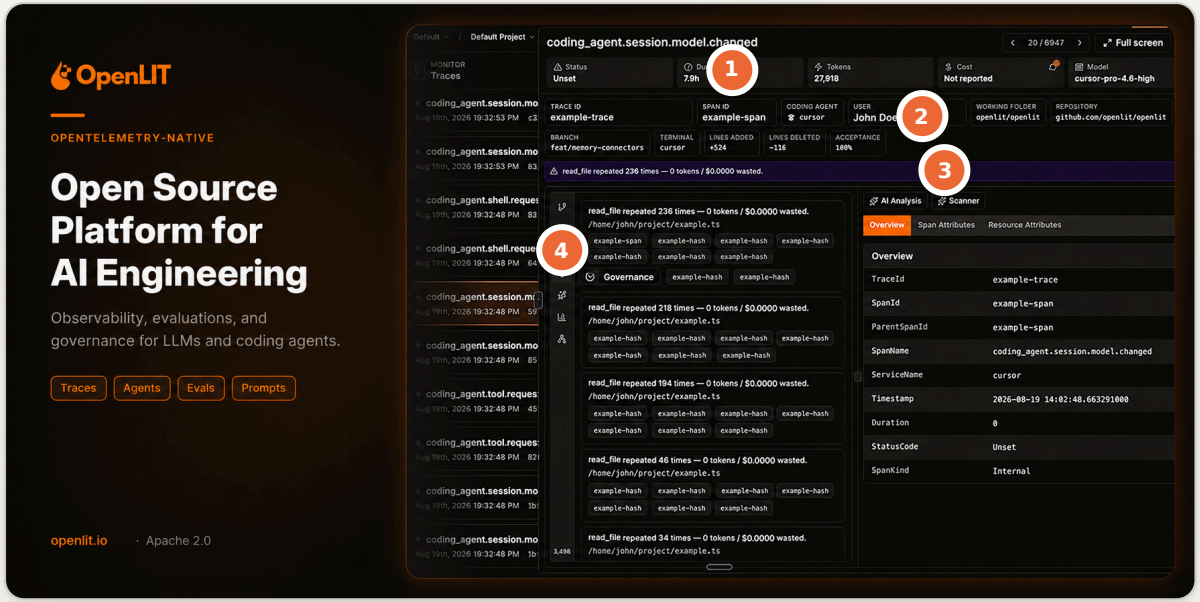
Tokens and cost by agent. Blind to processes.

S1 RERUN ✗	S2 WAITING ~	S3 POLLING ~	S4 LONG WAIT ~	S5 NO RESULT ~
Bash calls are listed in the transcript, but there is no CPU, memory or process data.	Lead and subagent are separate sessions with no link between them.	Counts of tool calls per session; repetition is found by reading.	Timestamps and session duration; no view of gaps. Waterfall needs OTLP traces.	Cost per session; outcome only through proxies: commits, PRs, lines changed.

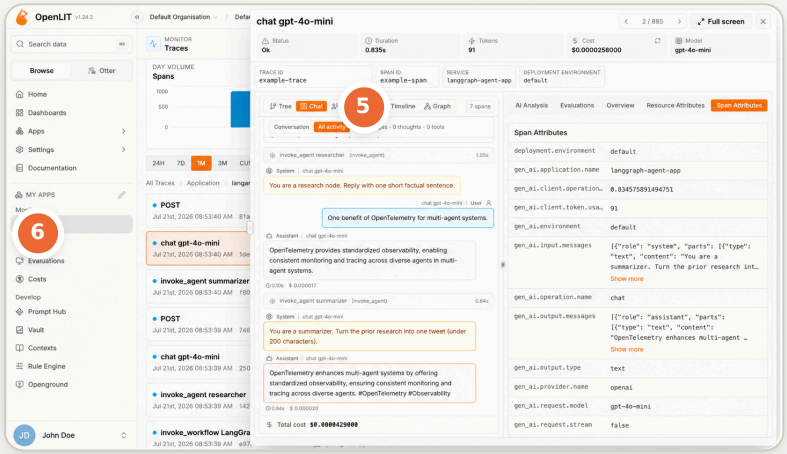
~ Tokens	✓ Cost	✗ Compute	~ Attribution	✓ Agents	✓ Setup
Input, cache read and write, output per agent. No per-model table.	Embedded prices for Claude, OpenAI, Gemini, Copilot.	CPU and memory only as Gemini CLI's own metrics.	Session → message → tool call. No subagent tree.	Claude Code, Codex, Gemini, Copilot; OpenCode via a third-party plugin.	One binary or container, local DuckDB. Already in Subyard.

ROLE **A light token and cost baseline. On its own it cannot show S1 or S2.**

Coding-agent sessions with loop badges.



Coding-agent trace. Public OpenLIT README example, anonymized.



Trace detail. Public OpenLIT docs example, anonymized.

1. **Run header**
   Status, duration, tokens, cost and model of the session.

2. **Coding context**
   Agent, user, working folder, repository, branch, terminal, lines added and deleted.

3. **Loop banner**
   “read_file repeated 236 times”, with wasted tokens and cost. Same tool, same arguments.

4. **Governance panel**
   Each repeated call group with its spans.

5. **Five views of a trace**
   Tree, Chat, AI Analysis, Timeline (waterfall over elapsed time) and Graph (DAG).

6. **Navigation**
   Telemetry, Agents, Evaluations, Costs, Prompt Hub, Vault, Rule Engine.

+ **Coding Agents → Sessions**
   Table with duration, model, tool calls, code changed, commits, PRs, tokens, cost and Outcome (merged, committed, abandoned, cancelled).

From documentation. Not run yet.

02 · OPENLIT FIT TO OUR GOALS

Closest to our scenarios. Compute lives next door.

S1 RERUN ~

The same test command twice gets a Loop badge. It compares arguments, not code; CPU is in a separate collector.

S2 WAITING ~

Subagent turns fold under the spawning chat; the timeline shows overlap. No “waiting” view.

S3 POLLING ✓

Repeated identical calls are flagged in the list and on the session row, with tokens and cost wasted.

S4 LONG WAIT ~

Wide spans in the timeline waterfall; waits are not labelled.

S5 NO RESULT ~

Cost and Outcome sit in the same table. Outcome is Git-based, not test results.

✓ Tokens
Per turn: input, output, cached, reasoning.

✓ Cost
Configurable price table, separate from billing.

~ Compute
Collector exports host and per-process CPU and memory, without a GPU too. Not joined to sessions.

✓ Attribution
Session → subagent (parent id) → tool span.

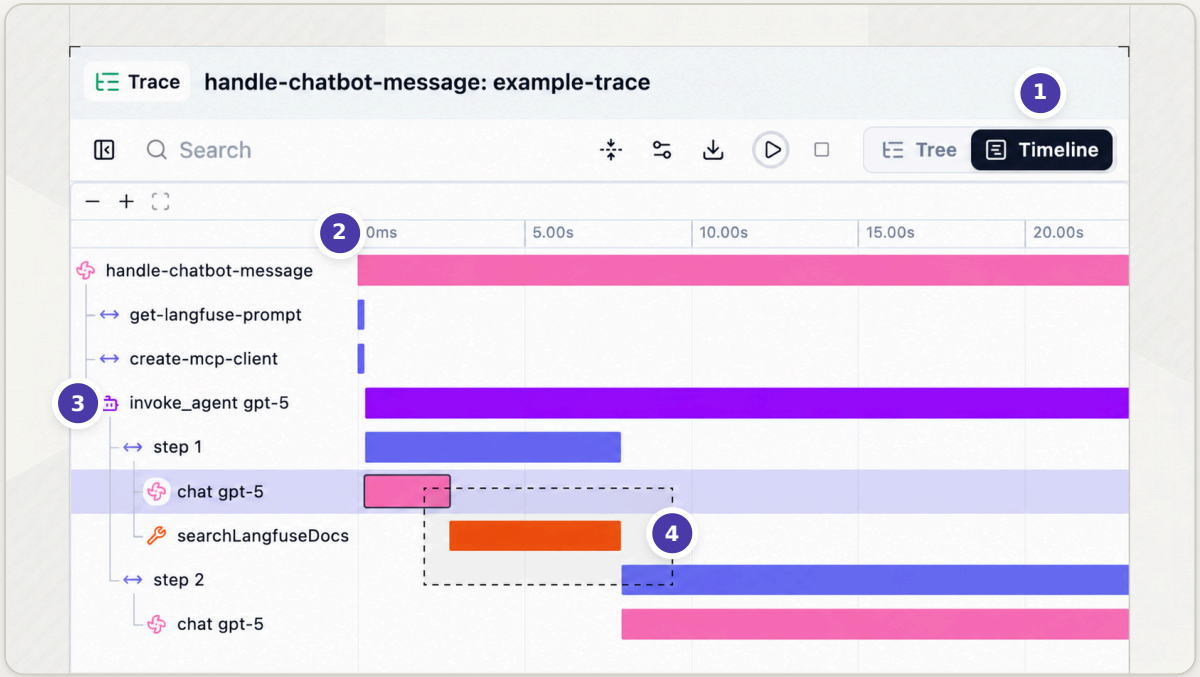
~ Agents
Claude Code, Codex, Cursor, Windsurf. No OpenCode.

✓ Setup
OpenLIT and ClickHouse containers plus a CLI. Content capture defaults to `full`.

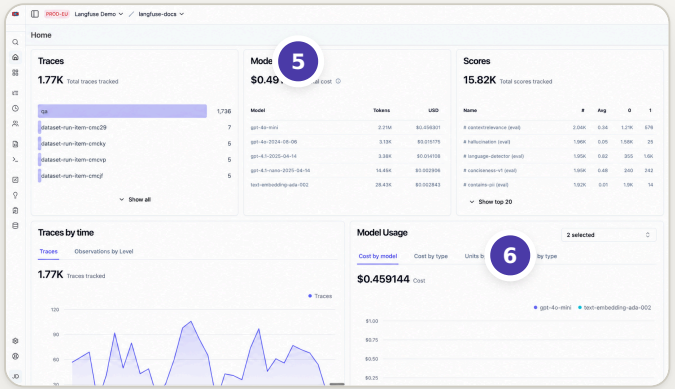
ROLE **Strongest agent-side fit on paper. Needs the process collector and one query to tie compute to a session.**

03 · LANGFUSE V4 INTERFACE

Traces you can read as a timeline.



Trace timeline, August 2026. Public example, anonymized.



Home dashboard. Public Langfuse docs example, anonymized.

1. **Tree or Timeline**
   Two views of one trace; the timeline zooms and pans from a dozen to over a thousand spans.
2. **Elapsed-time axis**
   Bar width is the duration of each observation.
3. **Nesting**
   Agent → step → model call or tool; colour marks the type.
4. **Tool call**
   A long tool call is a wide bar: the blocker is visible at a glance.
5. **Model costs**
   Tokens and USD per model.
6. **Model usage**
   Cost or units by model or by usage type.

+ **Agent graph**
  Aggregated mode merges repeated steps into one node with a counter and draws loops as cycles.

+ **Observations table**
  Default view in v4: every span with model, tokens, cost and latency; full-text search.

From documentation. Not run yet.

03 · LANGFUSE FIT TO OUR GOALS

A detailed view of the agent's steps. Nothing outside it.

S1 RERUN ~

Each test run is a tool span with its duration and command. No CPU, no code-change check.

S2 WAITING ~

Codex subagents nest under the spawning turn; repeated wait calls show as a counter.

S3 POLLING ~

The aggregated graph shows “step (n)”; no “no new information” check.

S4 LONG WAIT ✓

A long tool call is a wide bar. Waits between turns fall outside any trace.

S5 NO RESULT ~

Cost per trace and session; success needs a score that we define.

✓ **Tokens**
Usage types in separate buckets. Codex plugin: cached and reasoning.

✓ **Cost**
Model price table with overrides.

✗ **Compute**
Elapsed time per span only. No CPU, memory or processes.

✓ **Attribution**
Session → trace → generation → tool; subagents nested.

✓ **Agents**
Plugins for Claude Code, Codex and OpenCode.

~ **Setup**
Six services: web, worker, Postgres, ClickHouse, Redis, S3.

ROLE **Strong for S4 and attribution across all three agents. Needs a system-side partner for compute.**

04 · AGENTSIGHT V1.0.34 INTERFACE

Processes, files and LLM calls in one place.

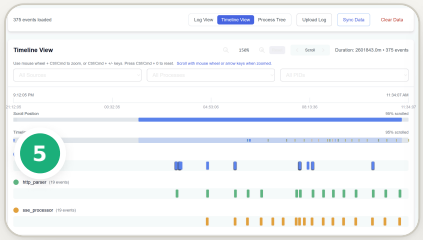
```
AgentSight top  mode=live sessions  state=running  refresh=2s  rows=10/10
sort cpu  view all  filter known agents  session tokens 1029.5m

sessions
  SESSION          AGENT    STATE   AGE    MODEL          TOKENS    HEALTH          ACTIVITY
> codex:rollou.0d  codex    live    42m    gpt-5.5        63.3M     398.6% 35.4G 7 proc, 714 tool, 10 file
  codex:rollou.cf  codex    live    6h     gpt-5.5        661.4M    7.5% 326M     4 proc, 6517 tool, 12 file
  codex:rollou.e6  codex    live    12h    gpt-5.5        55.1M     1.6% 396M     4 proc, 544 tool, 10 file
  claude:2f7b50.a  claude   live    11h    claude-opus-4  4.1M      0.9% 661M     4 proc, 94 tool, 1 file
  claude:d1fa99.a  claude   live    3h     claude-opus-4  118.6M    0.9% 513M     1 proc, 348 tool, 37 file
  claude:81d7b9.8  claude   live    40m    claude-opus-4  55.4k     0.6% 475M     3 proc, 1 tool

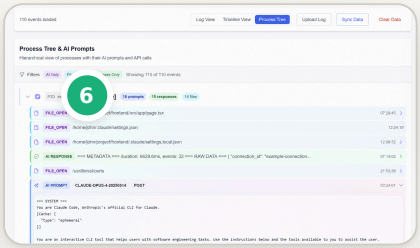
selected session - all
session: codex:rollou.0d3c5
agent: codex  state=live  pid=2307241
model: gpt-5.5

status
keys q quit | up/down select | s sort | v view | p pause | r refresh | +/- rows | ? help
session rows include agent-native local logs from ~/.claude or ~/.codex when present
proc evidence uses /proc for CPU/RSS/process families
```

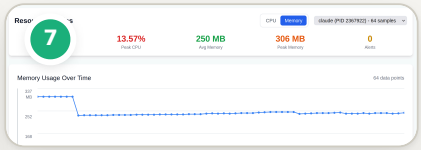
`agentsight top`, terminal. Official image.



Timeline



Process tree · anonymized



Metrics

1. **Live sessions, sorted by CPU**
   Refreshes every 2 s; works without root from `/proc` and session files.
2. **Tokens**
   Per session, with agent and model.
3. **CPU and memory**
   For the whole process family of the agent.
4. **Activity**
   Processes, tool calls and files touched.
5. **Timeline**
   One lane per source: processes, HTTP, model streams.
6. **Process tree**
   Each process with its prompts, responses and file opens.
7. **Metrics**
   Average and peak CPU and memory per process over time.

From documentation. Not run yet. `record -- <command>` saves a run to SQLite; `report` summarises it. The local tool is open source; a hosted app is in preview.

04 · AGENTSIGHT FIT TO OUR GOALS

CPU for the agent's processes. Weak on meaning.

S1 RERUN ~

Each test process shows with CPU and memory under the agent that started it. No code-change check.

S2 WAITING ~

LLM traffic next to a running child on the timeline; links between agents are not modelled.

S3 POLLING ~

README names repeated model and tool calls and friction signals; how they show is not documented.

S4 LONG WAIT ~

Gaps between events on the timeline; no wait label.

S5 NO RESULT ✗

Tokens and CPU are there; the outcome is not recorded.

~ Tokens

Input and output; cached and reasoning not documented.

✗ Cost

No price calculation documented.

✓ Compute

CPU and memory per process, process tree, files, network.

~ Attribution

Process → child process, not agent → subagent → tool.

✓ Agents

Any command; README names Claude Code, Codex, OpenCode.

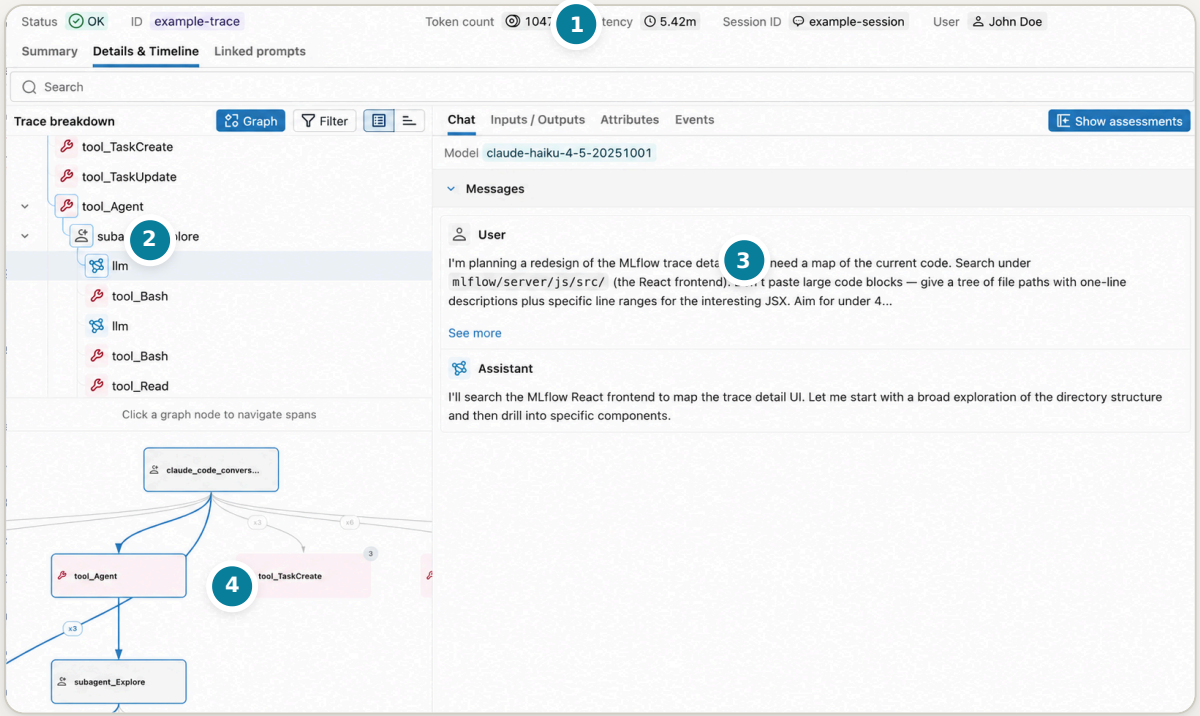
~ Setup

Linux only. eBPF needs root or `CAP_BPF`; in Docker, a privileged container with the host PID namespace.

ROLE **A compute partner for one of the agent-side tools, where root is available.**

05 · MLFLOW TRACE INTERFACE

Tools and nested subagents. One trace to inspect.



Claude Code trace. Public MLflow documentation example, anonymized.

1. **Trace summary**
   Status, tokens, elapsed time, session and user.

2. **Trace breakdown**
   Agent → subagent → model and tool calls. This screenshot shows Claude Code's nested activity.

3. **Inspect a step**
   Model and messages, inputs/outputs, attributes and events.

4. **Agent graph**
   Explore how tool and subagent spans relate.

+ **Details and Timeline**
  Inspect span durations to find a long recorded step.

+ **Three CLI integrations**
  Claude Code exports at session end; Codex after each turn; OpenCode when the session becomes idle.

From current documentation, 2026-10-12. Not run yet. Codex and OpenCode tool traces are documented; equivalent nested subagent linkage needs validation.

Usage over time. Keep cache accounting explicit.



Experiment Overview → Usage. Official Claude Code guide image.

1 Latency
Percentiles over the chosen time window reveal slow traces.

2 Errors
Execution error count and rate. An OK trace can still fail the development task.

3 Token buckets
Input, output, cache read and cache write are shown separately for this integration.

4 Tokens per trace
Distribution over time helps identify unusually large turns.

+ Estimated cost
Documented at span and trace level from model prices; requires server `mlflow[genai] ≥ 3.10`. Subscription spend can differ.

+ Quality and tool views
The Overview also has quality and tool-call tabs; outcome assessments must be supplied.

Accounting to validate: this example's headline excludes cache buckets. Reconcile provider fields so our input includes cached input once; reasoning stays part of output.

05 · MLFLOW FIT TO OUR GOALS

A local tracing and evaluation option. Compute needs glue.

S1 RERUN ~

Repeated tool spans and their duration. No code-change check or CPU per test process.

S2 WAITING ~

Claude subagents are nested. Relating lead calls to another agent's work needs inspection.

S3 POLLING ~

Read repeated calls and outputs; no documented check for new information.

S4 LONG WAIT ✓

Timeline and span durations expose long steps; gaps between turns need instrumentation.

S5 NO RESULT ~

Cost beside human or evaluated assessments. Task success is a definition we supply.

~ **Tokens**

Input/output plus provider-specific cache fields. Reasoning and totals need validation.

✓ **Cost**

Model-price estimates; supported model and usage required.

~ **Compute**

Opt-in host CPU/memory per MLflow run; no automatic CLI/process link.

~ **Attribution**

Session → tool/model span. Subagent coverage varies by integration.

✓ **Agents**

Official Claude Code, Codex and OpenCode integrations.

✓ **Setup**

Local server + SQLite + plugins. Disable usage telemetry for local-only use.

ROLE **Another agent-side candidate across all three CLIs. Pair with process observation for compute.**

No single tool covers it. Pairs might.

	AI Observer	OpenLIT	Langfuse	AgentSight	MLflow
SCENARIOS					
S1 Rerun without a change	✗ no process data	~ same-args loop	~ span duration	~ CPU per run	~ tool spans
S2 Expensive waiting	~ separate sessions	~ timeline overlap	~ nested subagents	~ process timeline	~ Claude nesting
S3 Polling	~ read transcript	✓ loop badge	~ graph counter	~ undocumented	~ read tool outputs
S4 Long wait	~ timestamps	~ waterfall	✓ timeline	~ event gaps	✓ span timeline
S5 Spend without a result	~ Git proxies	~ Git outcome	~ own scores	✗ no outcome	~ own assessments
REQUIREMENTS					
Tokens without double counting	~ per agent	✓ per turn	✓ per call	~ in / out	~ check buckets
Compute	✗	~ separate collector	✗	✓ per process	~ host metrics
Attribution	~ no subagents	✓	✓	~ process tree	~ varies by agent
Cost estimate	✓	✓	✓	✗	✓ model prices
Codex, Claude Code, OpenCode	✓	~ no OpenCode	✓	✓	✓ official plugins
Plain devcontainer	✓ one binary	✓ two containers	~ six services	~ needs root	✓ server + SQLite

✓ standard view ~ custom query or manual correlation ✗ data missing

AI Observer partly seen on a running instance; the rest is from documentation.

07 · OTHER OPTIONS

Other ways in, by the gap they fill.

Local log readers

ccusage · Sniffly

`ccusage claude|codex|opencode daily` prints token and cost tables from local session files, no server. Sniffly is a local web dashboard for Claude Code logs.

tokens · cost

Native OTel into a generic stack

Grafana LGTM · SigNoz

Send the agents' own OTLP to a general backend. Grafana Cloud has Claude Code and Codex integrations. S1–S5 dashboards are ours to build.

tokens · timeline

LLM gateway

LiteLLM · Portkey · Helicone

A proxy behind `ANTHROPIC_BASE_URL` logs tokens, cost and latency per request. It sees requests, not tools, subagents or processes.

tokens · cost

Other tracing platforms

W&B Weave · Opik · Phoenix · Laminar

Weave has a Claude Code plugin that traces turns, tool calls and subagents. Same class as Langfuse: no compute.

tokens · timeline

Process and container metrics

cAdvisor · Tetragon · Netdata

cAdvisor: CPU and memory per container. Tetragon: eBPF exec and exit events with the parent chain, so a test run traces back to the agent. Netdata: processes and cgroups.

compute

Reruns visible at the source

pytest cache · Nx · Turborepo · Bazel

Test and build caches key results on inputs. A cache hit, or a forced rerun, is a direct S1 signal that no observability tool infers today.

S1

Agent side plus system side. What stays open.

OpenLIT + its process collector

A. One UI, one database

Sessions with loop badges, outcome and cost; host and per-process CPU and memory in the same ClickHouse.

Open: tying process metrics to a session is a custom SQL dashboard. No OpenCode.

Langfuse or MLflow + AgentSight

B. Agent traces and processes

Agent-side traces and timelines; AgentSight for test processes, CPU and memory. MLflow offers a smaller local setup.

Open: two UIs correlated by time; eBPF needs root; MLflow's token buckets and subagent links need validation.

Native OTel + SigNoz or Grafana + cAdvisor or Tetragon

C. Configuration only

Claude Code traces and Codex OTel into a generic backend, with container or process metrics for compute.

Open: every S1–S5 view is a dashboard we write and maintain.

ccusage

D. A cross-check

Zero setup. Daily and per-session token and cost tables to validate the totals of whichever tool we pick.

Open: tokens and cost only.

NOT COVERED BY ANY **“No code or environment change since the last run” and “waiting for another agent” are not detected out of the box.**

Pick finalists. Try them on real runs.

AGENT SIDE · PROPOSAL

OpenLIT

Coding-agent sessions, loop detection, outcome column, and a process collector in the same stack.

SYSTEM SIDE · PROPOSAL

AgentSight

The only candidate that ties CPU, memory and test processes to an agent.

AGENT-SIDE ALTERNATIVES

Langfuse / MLflow

If OpenLIT falls short. Check MLflow's token accounting and subagent links on real runs.

01

Install

In an ordinary devcontainer next to Codex, Claude Code and OpenCode, without Subyard.

02

Use in real work

Everyday tasks, not synthetic ones (decision D7).

03

Note what was quick to see

For each weak spot S1–S5: which view showed it, and how long it took.

04

Record

Fill the ratings in `solutions/`, marked “tried”; log the choice in `docs/decisions.md`.